

Qualification Pack



Green Hydrogen Fuel Supply Chain Manager

QP Code: HYC/Q4006

Version: 1.0

NSQF Level: 6

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HYC/Q4006: Green Hydrogen Fuel Supply Chain Manager

Brief Job Description

Green hydrogen fuel Supply Chain Manager is in charge of every stage of an organizations supply chain, from purchasing raw materials to production of hydrogen. Individual is responsible for developing and monitoring of the companys supply chain strategy, keep track of logistics, inventory and ensure that the right product is available at the right time with a goal to improve productivity & efficiency and reduce costs, while securing high quality of green hydrogen that meet quality and safety standards.

Personal Attributes

The individual should have an excellent understanding of supply chain processes and good knowledge of relevant software and ERP systems. Overall, supply chain manager duties require high sense of responsibility, great project management and problem-solving skills awareness about financial transaction and good written and oral communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4001: Hydrogen Safety, Security and Health Management Procedures](#)
2. [HYC/N4011: Understanding basic concepts of Supply Chain Management](#)
3. [HYC/N4012: Managing Green Hydrogen Supply Chain](#)
4. [HYC/N4013: Roles and Responsibilities of Green hydrogen Supply Chain Manager](#)
5. [HYC/N9301: Working effectively in a team](#)
6. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
Country	India
NSQF Level	6

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Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO 2015/2113.0624
Minimum Educational Qualification & Experience	Completed 4 year UG program (BE/B.tech engineering in Chemical/Industrial/ Petrochemical / Mechanical branch) with 3 Years of experience OR Completed 3 year UG degree with 3-5 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	24 Years
Last Reviewed On	NA
Next Review Date	29/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	QG-06-HY-02525-2024-V1-HSSCI
NQR Version	1.0

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HYC/N4001: Hydrogen Safety, Security and Health Management Procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace where Hydrogen is produced and handled.

Scope

The scope covers the following :

- Understanding Properties of Hydrogen from safety point of view
- Safe handling of Hydrogen
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.
- PC2.** Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions
- PC3.** Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.
- PC4.** Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.
- PC5.** Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire
- PC6.** capability to accurately read and interpret data from sensing equipment used in hydrogen production processes
- PC7.** apply key safety parameters for handling hydrogen & oxygen
- PC8.** Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames
- PC9.** Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards
- PC10.** competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace
- PC11.** Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations
- PC12.** Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.

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- PC13.** safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries
- PC14.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC15.** Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.
- PC16.** Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.
- PC17.** Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.

Ensure the following fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC18.** Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly
- PC19.** Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard
- PC20.** Promote and enforce following good housekeeping practice in order to prevent fire hazards
- PC21.** Regularly inspect work areas for compliance with housekeeping standards
- PC22.** identify and document issues concerning the hydrogen safety in work place
- PC23.** Collaborate with relevant departments to address and rectify safety concerns promptly
- PC24.** Ensure that work areas and escape routes are well maintained and free from any hindrance
- PC25.** Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation
- PC26.** Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard
- PC27.** Ensure that personnel are trained in preparing comprehensive accident/incident reports
- PC28.** Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC29.** provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues
- PC30.** Be proficient in basic techniques of bandaging
- PC31.** Lead by example in responding promptly and appropriately to an accident
- PC32.** perform rescue activity during an accident in real or simulated environments
- PC33.** Familiarize personnel with correct escape routes during an emergency
- PC34.** demonstrate correct method to rescue injured people and others during an emergency
- PC35.** Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance
- KU12.** the concept of Mission Karmayogi
- KU13.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen</i>	17	27	-	-
PC1. Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.	1	2	-	-
PC2. Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions	1	1	-	-
PC3. Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.	1	2	-	-
PC4. Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.	1	2	-	-
PC5. Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire	1	2	-	-
PC6. capability to accurately read and interpret data from sensing equipment used in hydrogen production processes	1	2	-	-
PC7. apply key safety parameters for handling hydrogen & oxygen	1	2	-	-
PC8. Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames	1	1	-	-
PC9. Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards	1	1	-	-
PC10. competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations	1	1	-	-
PC12. Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.	1	1	-	-
PC13. safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries	1	2	-	-
PC14. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC15. Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.	1	2	-	-
PC16. Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.	1	1	-	-
PC17. Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.	1	1	-	-
<i>Ensure the following fire safety procedures</i>	11	19	-	-
PC18. Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly	1	2	-	-
PC19. Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard	1	2	-	-
PC20. Promote and enforce following good housekeeping practice in order to prevent fire hazards	1	2	-	-
PC21. Regularly inspect work areas for compliance with housekeeping standards	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify and document issues concerning the hydrogen safety in work place	1	2	-	-
PC23. Collaborate with relevant departments to address and rectify safety concerns promptly	1	2	-	-
PC24. Ensure that work areas and escape routes are well maintained and free from any hindrance	1	2	-	-
PC25. Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation	1	2	-	-
PC26. Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard	1	1	-	-
PC27. Ensure that personnel are trained in preparing comprehensive accident/incident reports	1	1	-	-
PC28. Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions	1	1	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	7	14	-	-
PC29. provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues	1	2	-	-
PC30. Be proficient in basic techniques of bandaging	1	2	-	-
PC31. Lead by example in responding promptly and appropriately to an accident	1	2	-	-
PC32. perform rescue activity during an accident in real or simulated environments	1	2	-	-
PC33. Familiarize personnel with correct escape routes during an emergency	1	2	-	-
PC34. demonstrate correct method to rescue injured people and others during an emergency	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.	1	2	-	-
NOS Total	35	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4001
NOS Name	Hydrogen Safety, Security and Health Management Procedures
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4011: Understanding basic concepts of Supply Chain Management

Description

This unit is about introducing basic concept of supply chain management and their significance for enhancing operational efficiency and profitability.

Scope

The scope covers the following :

- Supply chain performance, drivers and metrics
- Planning and coordinating demand and supply
- Sales and operations planning
- Managing uncertainties in supply chain

Elements and Performance Criteria

Supply chain performance, drivers and metrics

To be competent, the user/individual on the job must be able to:

- PC1.** Understand what is a supply chain.
- PC2.** Know recent trend of supply chain
- PC3.** Comprehend role of analytics in supply chain
- PC4.** Identify supply chain strategies push-pull chain
- PC5.** Appreciate importance of Supply chain strategies w.r.t efficiency and responsiveness
- PC6.** Understand the customer and supply chain uncertainty
- PC7.** Know tools/drivers for supply chain management
- PC8.** Design supply chain or distribution network
- PC9.** Identify parameters to decide suitable distribution
- PC10.** optimize plants and warehouses simultaneously
- PC11.** Design global Supply Chain Network

Planning and coordinating demand and supply

To be competent, the user/individual on the job must be able to:

- PC12.** Understand basics of demand forecasting
- PC13.** Know time series analysis for forecasting
- PC14.** Identify forecasting errors, tracking signal, Sigma
- PC15.** use of Python for forecasting

Sales and operations planning

To be competent, the user/individual on the job must be able to:

- PC16.** Plan supply and demand in supply chain
- PC17.** Identify cost of overstocking and understocking
- PC18.** optimize inventory separately for two installations
- PC19.** Understand role of aggregate planning in supply chain

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- PC20.** Know how to do aggregate planning
- PC21.** Identify and respond to predictable variability in supply chain
- PC22.** Know implications of lack of coordination in supply chain
- PC23.** Understand short term discounting for business profitability

Managing uncertainties in supply chain

To be competent, the user/individual on the job must be able to:

- PC24.** Understand impact of lack of supply chain coordination and Bullwhip effect
- PC25.** Manage uncertainties in supply chain
- PC26.** Identify optimal cycle service level
- PC27.** Identify Impact of supply uncertainties on safety inventory
- PC28.** Know how to improve supply chain profitability
- PC29.** Grasp importance of transportation in Supply Chain and its Impact on profit and inventories
- PC30.** Identify future Issues in supply chain

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** How to manage huge amount of data
- KU2.** Usage of Microsoft excel
- KU3.** Supply change in different industries and segments
- KU4.** The procurement processes of the organisation
- KU5.** software to manage logistical tasks like procurement, inventory management and other supply chain planning and management processes.
- KU6.** the manufacturing process in green hydrogen industry
- KU7.** Various transportation methods employed for green hydrogen
- KU8.** Requirement of storage capacities in the green hydrogen plant
- KU9.** Risk mitigation strategies of the organisation
- KU10.** Tools like artificial intelligence, machine learning, Internet of Things, analytics softwares
- KU11.** Existing vendor base of the organisation
- KU12.** alternate vendors in case of natural disaster
- KU13.** ability to use data to track orders and shipments
- KU14.** sales trends, demand and supply issues of green hydrogen
- KU15.** Big-picture about global green hydrogen landscape

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate effectively
- GS2.** share the information efficiently
- GS3.** use of computers (MS office tools)

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- GS4.** Presentation skills -oral and written
- GS5.** Effectively use, read and interpret the data
- GS6.** read instructions, guidelines/procedures and reports
- GS7.** analyze the parameters and output
- GS8.** Liaison with suppliers and build relationship
- GS9.** Solve a problem

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Supply chain performance, drivers and metrics</i>	32	21	-	-
PC1. Understand what is a supply chain.	3	2	-	-
PC2. Know recent trend of supply chain	2	1	-	-
PC3. Comprehend role of analytics in supply chain	3	2	-	-
PC4. Identify supply chain strategies push-pull chain	3	2	-	-
PC5. Appreciate importance of Supply chain strategies w.r.t efficiency and responsiveness	3	2	-	-
PC6. Understand the customer and supply chain uncertainty	3	2	-	-
PC7. Know tools/drivers for supply chain management	3	2	-	-
PC8. Design supply chain or distribution network	3	2	-	-
PC9. Identify parameters to decide suitable distribution	3	2	-	-
PC10. optimize plants and warehouses simultaneously	3	2	-	-
PC11. Design global Supply Chain Network	3	2	-	-
<i>Planning and coordinating demand and supply</i>	8	4	-	-
PC12. Understand basics of demand forecasting	2	1	-	-
PC13. Know time series analysis for forecasting	2	1	-	-
PC14. Identify forecasting errors, tracking signal, Sigma	2	1	-	-
PC15. use of Python for forecasting	2	1	-	-
<i>Sales and operations planning</i>	16	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. Plan supply and demand in supply chain	2	1	-	-
PC17. Identify cost of overstocking and understocking	2	1	-	-
PC18. optimize inventory separately for two installations	2	1	-	-
PC19. Understand role of aggregate planning in supply chain	2	1	-	-
PC20. Know how to do aggregate planning	2	1	-	-
PC21. Identify and respond to predictable variability in supply chain	2	1	-	-
PC22. Know implications of lack of coordination in supply chain	2	1	-	-
PC23. Understand short term discounting for business profitability	2	1	-	-
<i>Managing uncertainties in supply chain</i>	14	7	-	-
PC24. Understand impact of lack of supply chain coordination and Bullwhip effect	2	1	-	-
PC25. Manage uncertainties in supply chain	2	1	-	-
PC26. Identify optimal cycle service level	2	1	-	-
PC27. Identify Impact of supply uncertainties on safety inventory	2	1	-	-
PC28. Know how to improve supply chain profitability	2	1	-	-
PC29. Grasp importance of transportation in Supply Chain and its Impact on profit and inventories	2	1	-	-
PC30. Identify future Issues in supply chain	2	1	-	-
NOS Total	70	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4011
NOS Name	Understanding basic concepts of Supply Chain Management
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4012: Managing Green Hydrogen Supply Chain

Description

This unit is about the green hydrogen supply chain, various challenges associated with it and mitigation of the challenges.

Scope

The scope covers the following :

- Working of green hydrogen value chain
- Green hydrogen production, storage, transportation and applications

Elements and Performance Criteria

Working of green hydrogen value chain

To be competent, the user/individual on the job must be able to:

- PC1.** Understand what is the hydrogen economy
- PC2.** Identify different green hydrogen supply chains
- PC3.** Understand global hydrogen supply chain
- PC4.** Know about Green hydrogen supply chain
- PC5.** Understand how green hydrogen produced at industrial scale
- PC6.** Identify challenges and comparison of different methods of green hydrogen production
- PC7.** Work out Hydrogen production cost
- PC8.** Know green hydrogen electrolyzers supply chain
- PC9.** Understand different hydrogen storage methods
- PC10.** Identify challenges and comparison of different methods of hydrogen storage
- PC11.** Understand different ways of hydrogen transportation
- PC12.** Identify challenges and comparison of different ways of hydrogen transportation
- PC13.** Understand different applications of green hydrogen
- PC14.** Identify challenges of supply chain in different applications
- PC15.** Know demand supply scenario of the product where green hydrogen is used
- PC16.** Track upcoming green hydrogen projects and their impact on demand and supply
- PC17.** Follow growth of hydrogen fuel cell vehicles
- PC18.** The Importance of Location in the Green Hydrogen Supply Chain
- PC19.** Identify challenges in Building a Green Hydrogen Supply Chain
- PC20.** Understand Evolution and future of hydrogen supply chain
- PC21.** Technical Challenges and Current Limitations
- PC22.** Infrastructure Requirements for Green Hydrogen Supply Chains
- PC23.** Track the upcoming solar projects and capacities
- PC24.** Track the upcoming wind projects and capacities

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- PC25.** Follow the integration of renewable power with grid
- PC26.** Identify Hydrogen supply chain companies
- PC27.** Comprehend role of Government Policies in Promoting Green Hydrogen
- PC28.** Understand policy and Regulatory Challenges
- PC29.** Predict green hydrogen supply - demand scenario
- PC30.** Reduce cost at every level
- PC31.** Manage the teams and lead the company strategy toward more efficient, cost-effective production timelines
- PC32.** Manage the manufacturing and distribution processes
- PC33.** Coordinate with procurement managers, buyers, and vendors to source the right materials
- PC34.** Negotiate contracts with suppliers, vendors, and customers
- PC35.** Utilizing software to track goods from factory to warehouse
- PC36.** Employ data analytics to forecast demand for inventory
- PC37.** analyze performance of products
- PC38.** Developing new and existing relationships with suppliers and partners
- PC39.** Innovating the supply chain process and efficiency
- PC40.** Stay abreast of trends and developments in the sector and technologies

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamental of green hydrogen supply chain
- KU2.** Green hydrogen production technology employed in the organisation
- KU3.** Green hydrogen storage technologies employed in the organisation
- KU4.** Available green hydrogen transportation infrastructure
- KU5.** Client requirement in terms of end applications
- KU6.** Usage of Microsoft excel
- KU7.** Supply change in different industries and segments
- KU8.** The procurement processes of the organisation
- KU9.** software to manage logistical tasks like procurement, inventory management and other supply chain planning and management processes.
- KU10.** Risk mitigation strategies of the organisation
- KU11.** Tools like artificial intelligence, machine learning, Internet of Things, analytics softwares
- KU12.** Existing vendor base of the organisation
- KU13.** alternate vendors in case of natural disaster
- KU14.** ability to use data to track orders and shipments
- KU15.** sales trends, demand and supply issues of green hydrogen
- KU16.** Big-picture about global green hydrogen landscape
- KU17.** The concept of Mission Karmayogi.
- KU18.** The concept of Swachhata hi Seva.

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate effectively
- GS2.** share the information efficiently
- GS3.** use of computers (MS office tools)
- GS4.** Presentation skills -oral and written.
- GS5.** Effectively use, read and interpret the data
- GS6.** read instructions, guidelines/procedures and reports
- GS7.** analyze the parameters and output
- GS8.** Liaison with suppliers and build relationship
- GS9.** Solve a problem

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Working of green hydrogen value chain</i>	80	40	-	-
PC1. Understand what is the hydrogen economy	2	1	-	-
PC2. Identify different green hydrogen supply chains	2	1	-	-
PC3. Understand global hydrogen supply chain	2	1	-	-
PC4. Know about Green hydrogen supply chain	2	1	-	-
PC5. Understand how green hydrogen produced at industrial scale	2	1	-	-
PC6. Identify challenges and comparison of different methods of green hydrogen production	2	1	-	-
PC7. Work out Hydrogen production cost	2	1	-	-
PC8. Know green hydrogen electrolyzers supply chain	2	1	-	-
PC9. Understand different hydrogen storage methods	2	1	-	-
PC10. Identify challenges and comparison of different methods of hydrogen storage	2	1	-	-
PC11. Understand different ways of hydrogen transportation	2	1	-	-
PC12. Identify challenges and comparison of different ways of hydrogen transportation	2	1	-	-
PC13. Understand different applications of green hydrogen	2	1	-	-
PC14. Identify challenges of supply chain in different applications	2	1	-	-
PC15. Know demand supply scenario of the product where green hydrogen is used	2	1	-	-
PC16. Track upcoming green hydrogen projects and their impact on demand and supply	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC17. Follow growth of hydrogen fuel cell vehicles	2	1	-	-
PC18. The Importance of Location in the Green Hydrogen Supply Chain	2	1	-	-
PC19. Identify challenges in Building a Green Hydrogen Supply Chain	2	1	-	-
PC20. Understand Evolution and future of hydrogen supply chain	2	1	-	-
PC21. Technical Challenges and Current Limitations	2	1	-	-
PC22. Infrastructure Requirements for Green Hydrogen Supply Chains	2	1	-	-
PC23. Track the upcoming solar projects and capacities	2	1	-	-
PC24. Track the upcoming wind projects and capacities	2	1	-	-
PC25. Follow the integration of renewable power with grid	2	1	-	-
PC26. Identify Hydrogen supply chain companies	2	1	-	-
PC27. Comprehend role of Government Policies in Promoting Green Hydrogen	2	1	-	-
PC28. Understand policy and Regulatory Challenges	2	1	-	-
PC29. Predict green hydrogen supply - demand scenario	2	1	-	-
PC30. Reduce cost at every level	2	1	-	-
PC31. Manage the teams and lead the company strategy toward more efficient, cost-effective production timelines	2	1	-	-
PC32. Manage the manufacturing and distribution processes	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. Coordinate with procurement managers, buyers, and vendors to source the right materials	2	1	-	-
PC34. Negotiate contracts with suppliers, vendors, and customers	2	1	-	-
PC35. Utilizing software to track goods from factory to warehouse	2	1	-	-
PC36. Employ data analytics to forecast demand for inventory	2	1	-	-
PC37. analyze performance of products	2	1	-	-
PC38. Developing new and existing relationships with suppliers and partners	2	1	-	-
PC39. Innovating the supply chain process and efficiency	2	1	-	-
PC40. Stay abreast of trends and developments in the sector and technologies	2	1	-	-
NOS Total	80	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4012
NOS Name	Managing Green Hydrogen Supply Chain
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	6
Credits	5
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4013: Roles and Responsibilities of Green hydrogen Supply Chain Manager

Description

This unit covers major roles and responsibilities of a green hydrogen supply chain manager

Scope

The scope covers the following :

- Expected role of a green hydrogen supply chain manager engaged by different industries operating in the business of green hydrogen.

Elements and Performance Criteria

Roles and Responsibilities

To be competent, the user/individual on the job must be able to:

- PC1.** Manage the teams and lead the strategy toward more efficient, cost-effective production timelines for green hydrogen
- PC2.** Manage the manufacturing and distribution processes
- PC3.** Coordinate with procurement managers, buyers, and vendors to source the right materials or products
- PC4.** keep track of logistics and update the company's inventory
- PC5.** Negotiate contracts with suppliers, vendors, contractors, and customers
- PC6.** Utilize software to track goods from factory to warehouse
- PC7.** Use data analytics to forecast demand for inventory
- PC8.** analyze performance of products, and measure against the economy (and other factors)
- PC9.** Optimise and reduce costs at every level while maintaining quality and meeting sustainability targets
- PC10.** Develop relationships with new and existing relationships with suppliers and partners
- PC11.** Continuously innovate the supply chain process and efficiency
- PC12.** Stay abreast of trends and developments in the sector and technologies
- PC13.** Analyse project, supply chain management for timely execution
- PC14.** Regularly forecasts for purchased supply (project-wise)
- PC15.** Maintain vendor database management, negotiation, and conduct regular visits
- PC16.** Review reports for deliverables and expedite open orders as required
- PC17.** participate in all the aspects of business that have a bearing on supply chains
- PC18.** Understand sales forecasting
- PC19.** Know quality management
- PC20.** Provide customer services
- PC21.** Track changes in production and business climate that affect supply chains
- PC22.** create alternative supply chains as the need arises Evaluate and report on KPIs

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- PC23.** Monitor logistics to make sure they run smoothly
- PC24.** Maintain supply chain inventory and records
- PC25.** Train and guide employees
- PC26.** Find cost-effective solutions for supply chain processes
- PC27.** Resolve issues that come up (e.g. delays in delivery, accidents)
- PC28.** Collaborate with other departments to create coordinated plans for business growth
- PC29.** Develop and implement safety guidelines in all aspects of the supply chain (e.g. use of trucks, forklifts)
- PC30.** Ensure supply chain processes meet legal requirements and standards
- PC31.** Communicate and negotiate with suppliers and vendors to land more profitable deals

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamental of green hydrogen supply chain
- KU2.** Green hydrogen production technology employed in the organisation
- KU3.** Green hydrogen storage technologies employed in the organisation
- KU4.** Available green hydrogen transportation infrastructure
- KU5.** Client requirement in terms of end applications
- KU6.** Usage of Microsoft excel
- KU7.** Supply change in different industries and segments
- KU8.** The procurement processes of the organisation
- KU9.** software to manage logistical tasks like procurement, inventory management and other supply chain planning and management processes.
- KU10.** Risk mitigation strategies of the organisation
- KU11.** Tools like artificial intelligence, machine learning, Internet of Things, analytics softwares
- KU12.** Existing vendor base of the organisation
- KU13.** alternate vendors in case of natural disaster
- KU14.** ability to use data to track orders and shipments,
- KU15.** sales trends, demand and supply issues of green hydrogen
- KU16.** Big-picture about global green hydrogen landscape
- KU17.** The concept of Mission Karmayogi.
- KU18.** The concept of Swachhata hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Work out Costing
- GS2.** Negotiate with Supplier
- GS3.** Carry out contract negotiation

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- GS4.** Carry out procurement as per company policy
- GS5.** Implement process Improvement
- GS6.** Apply logistics skills like shipping and trucks routs
- GS7.** Apply previous experience as a supply chain manager in a similar field
- GS8.** Use relevant software (e.g. SAP)
- GS9.** communication effectively and clearly
- GS10.** Give attention to detail
- GS11.** Solve a problem creatively

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Roles and Responsibilities</i>	60	30	-	-
PC1. Manage the teams and lead the strategy toward more efficient, cost-effective production timelines for green hydrogen	2	1	-	-
PC2. Manage the manufacturing and distribution processes	2	1	-	-
PC3. Coordinate with procurement managers, buyers, and vendors to source the right materials or products	2	1	-	-
PC4. keep track of logistics and update the company's inventory	2	1	-	-
PC5. Negotiate contracts with suppliers, vendors, contractors, and customers	2	1	-	-
PC6. Utilize software to track goods from factory to warehouse	2	1	-	-
PC7. Use data analytics to forecast demand for inventory	2	1	-	-
PC8. analyze performance of products, and measure against the economy (and other factors)	2	1	-	-
PC9. Optimise and reduce costs at every level while maintaining quality and meeting sustainability targets	2	1	-	-
PC10. Develop relationships with new and existing relationships with suppliers and partners	2	1	-	-
PC11. Continuously innovate the supply chain process and efficiency	2	1	-	-
PC12. Stay abreast of trends and developments in the sector and technologies	2	1	-	-
PC13. Analyse project, supply chain management for timely execution	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Regularly forecasts for purchased supply (project-wise)	2	1	-	-
PC15. Maintain vendor database management, negotiation, and conduct regular visits	2	1	-	-
PC16. Review reports for deliverables and expedite open orders as required	2	1	-	-
PC17. participate in all the aspects of business that have a bearing on supply chains	2	1	-	-
PC18. Understand sales forecasting	2	-	-	-
PC19. Know quality management	1	1	-	-
PC20. Provide customer services	1	1	-	-
PC21. Track changes in production and business climate that affect supply chains	2	1	-	-
PC22. create alternative supply chains as the need arises Evaluate and report on KPIs	2	1	-	-
PC23. Monitor logistics to make sure they run smoothly	2	1	-	-
PC24. Maintain supply chain inventory and records	2	1	-	-
PC25. Train and guide employees	2	1	-	-
PC26. Find cost-effective solutions for supply chain processes	2	1	-	-
PC27. Resolve issues that come up (e.g. delays in delivery, accidents)	2	1	-	-
PC28. Collaborate with other departments to create coordinated plans for business growth	2	1	-	-
PC29. Develop and implement safety guidelines in all aspects of the supply chain (e.g. use of trucks, forklifts)	2	1	-	-
PC30. Ensure supply chain processes meet legal requirements and standards	2	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC31. Communicate and negotiate with suppliers and vendors to land more profitable deals	2	1	-	-
NOS Total	60	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4013
NOS Name	Roles and Responsibilities of Green hydrogen Supply Chain Manager
Sector	Hydrocarbon
Sub-Sector	
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	6
Credits	3.5
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

Qualification Pack

HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2.5
Version	5.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training centre (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4001.Hydrogen Safety, Security and Health Management Procedures	35	60	-	-	95	30
HYC/N4011.Understanding basic concepts of Supply Chain Management	70	40	0	0	110	30
HYC/N4012.Managing Green Hydrogen Supply Chain	80	40	0	0	120	20
HYC/N4013.Roles and Responsibilities of Green hydrogen Supply Chain Manager	60	30	0	0	90	10
HYC/N9301.Working effectively in a team	20	30	-	-	50	5
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	5
Total	285	230	0	0	515	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure

Qualification Pack

NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard Operating Procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding

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GS	Generic Skills
SOP	Standard Operating Procedure

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.